Part IV Laplace Transform

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4-1 The Laplace Transform

$$X(s) = \int_{0^-}^{\infty} x(t)e^{-st}\mathrm{d}t$$

which is called as the one-sided Laplace Transform

4-2 Useful Equations and Properties Table

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) \longleftrightarrow sX(s) - x(0^-)$$
$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) \longleftrightarrow s^2X(s) - sx(0^-) - x'(0^-)$$

Aperiodic Signal	Fourier Signal	Signal	Transform	ROC
$ax(t)+by(t)$	$aX(s)+bY(s)$	$\delta(t)$	1	All s
$x(t-t_0)$	$e^{-st_0}X(s)$	$u(t)$	$\frac{1}{s}$	$s>0$
$e^{s_0t}x(t)$	$X(s-s_0)$	$-u(-t)$	$\frac{1}{s}$	$s<0$
$x^*(t)$	$X^*(s^*)$	$\frac{t^{n-1}}{(n-1)!}u(t)$	$\frac{1}{s^n}$	$s>0$
$x(at)$	$\frac{1}{\lvert a\rvert}X(\frac{s}{a})$	$-\frac{t^{n-1}}{(n-1)!}u(-t)$	$\frac{1}{s^n}$	$s<0$
$x(-t)$	$X(-s)$	$e^{-\alpha t}u(t)$	$\frac{1}{s+\alpha}$	$s>-\alpha$
$x(t)*y(t)$	$X(s)Y(s)$	$-e^{-\alpha t}u(-t)$	$\frac{1}{s+\alpha}$	$s<-\alpha$
$\frac{\mathrm{d}}{\mathrm{d}t}x(t)$	$sX(s)$	$\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(t)$	$\frac{1}{(s+\alpha)^n}$	$s>-\alpha$
$\int_{-\infty}^{t}x(t)\mathrm{d}t$	$\frac{1}{s}X(s)$	$-\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(-t)$	$\frac{1}{(s+\alpha)^n}$	$s<-\alpha$
$-tx(t)$	$\frac{\mathrm{d}}{\mathrm{d}s}X(s)$	$\delta(t-T)$	e^{-sT}	All s
		$[\cos\omega_0t]u(t)$	$\frac{s}{s^2+\omega_0^2}$	$s>0$
		$[\sin\omega_0t]u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$s>0$
		$[e^{-\alpha t}\cos\omega_0t]u(t)$	$\frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$	$s>\alpha$
		$[e^{-\alpha t}\sin\omega_0t]u(t)$	$\frac{\omega_0}{(s+\alpha)^2+\omega_0^2}$	$s>\alpha$

4-3 The Inverse Laplace Transform

Given $F(s)$, how do we transform it back to the time domain and obtain the corresponding $f(t)$

$$X(s)=\frac{N(s)}{D(s)}$$

- the roots of $N(s)$ are defined as the zeros

- the roots of $D(s)$ are defined as the poles

$F(s)$ can always be reduced to a ratio with $m < n$ by polynomial division

Simple Poles

if the degree of $N(s)$ is less than the degree of $D(s)$

$$F(s) = \frac{k_1}{s+p_1} + \frac{k_2}{s+p_2} + \cdots + \frac{k_n}{s+p_n}$$

> where $k_i = (s+p_i)F(s)\Big|_{s=-p_i}$

Repeated Poles

$$F(s) = \frac{k_n}{(s+p)^n} + \frac{k_{n-1}}{(s+p)^{n-1}} + \cdots + \frac{k_1}{s+p}$$

Complex Poles

$$F(s) = \frac{A_1 s + A_2}{s^2 + as + b} = \frac{A_1(s+\alpha)}{(s+\alpha)^2+\beta^2} + \frac{B_1\beta}{(s+\alpha)^2+\beta^2}$$

4-4 Initial and Final Value Theorems

- Initial Value: $f(0) = \lim_{s\to\infty} sF(s)$
- Final Value: $f(\infty) = \lim_{s\to 0} sF(s)$

> Final value theorem **fails** if the real part of pole is **non-negative**

4-5 The Convolution Integral

$$y(t) = \int x(\tau)h(t-\tau)\mathrm{d}\tau \overset{L}{\Longleftrightarrow} Y(s) = X(s)H(s)$$

4-6 Applications of Laplace Transform

- Resistor: $Z(s) = R$
- Inductor: $Z(s) = sL$
- Capacitor: $Z(s) = \frac{1}{sC}$